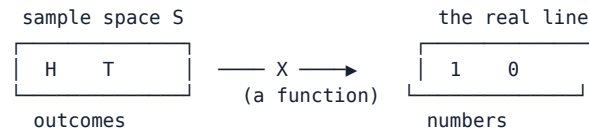


02 RANDOM VARIABLES (notes p051-p062)

1. What a random variable IS (p051)

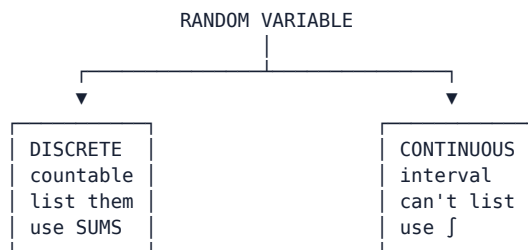
A random variable is a FUNCTION whose domain is the sample space and whose range is the set of real numbers.

It is NOT the outcome. It is the NUMBER you attach to it.



$$X(H) = 1$$

$$X(T) = 0$$



2. Example 1: Bernoulli (p052)

toss one fair coin, $S = \{H, T\}$
define $X(H) = 1$, $X(T) = 0$

DEFINITION: any rv whose only values are 0 and 1 is a BERNOLLI random variable.

Bernoulli is the atom of the binomial: one trial, success = 1, failure = 0.

3. Example 2: two coins (p053)

$S = \{HH, HT, TH, TT\}$
 $X = \text{number of heads}$

outcome	X value
HH	2
HT	1
TH	1
TT	0

$$X(HH)=2, \quad X(TT)=0, \quad X(TH)=1, \quad X(HT)=1$$

Notice HT and TH give the SAME X. Different outcomes, same value. That collapse is how a pmf is born (next file).

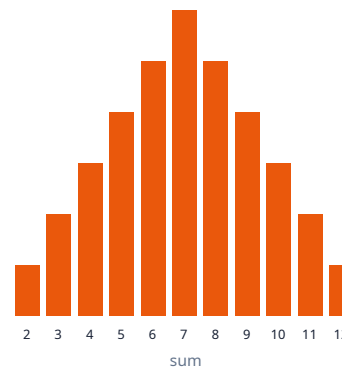
4. Example 3: dice sum (p054-p055)

Two dice: 36 ordered outcomes

	1	2	3	4	5	6
die 1						
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

cell = ordered pair; colour = sum

sum frequencies

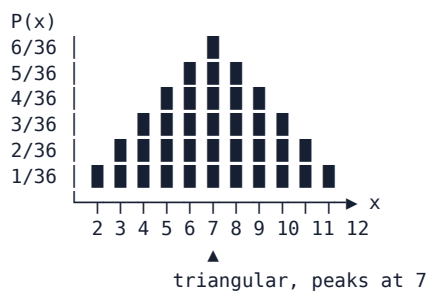


Ordered outcomes and the distribution of their sum

X = sum of two fair dice

x	outcomes producing it	count	P(X=x)
2	(1,1)	1	1/36
3	(1,2) (2,1)	2	2/36
4	(1,3) (2,2) (3,1)	3	3/36
5	(1,4) (2,3) (3,2) (4,1)	4	4/36
6	5 outcomes	5	5/36
7	6 outcomes	6	6/36
8	5 outcomes	5	5/36
9	4 outcomes	4	4/36
10	3 outcomes	3	3/36
11	(5,6) (6,5)	2	2/36
12	(6,6)	1	1/36
	Σ	36/36 = 1	✓

The pmf as a picture:



The identity the slide writes:

$$1 = P\left(\bigcup_{n=2}^{12} (X=n)\right) = \sum_{n=2}^{12} P(X=n)$$

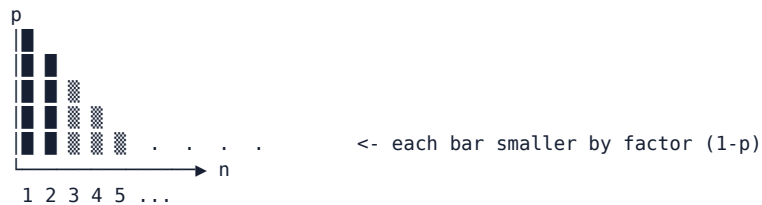
5. Example 4: two coins as Y = heads (p056)

$$\begin{aligned} P(Y=0) &= P(TT) &= 1/4 \\ P(Y=1) &= P(HT, TH) &= 2/4 \\ P(Y=2) &= P(HH) &= 1/4 \\ \text{sum} &= 1 \quad \checkmark \end{aligned}$$

6. Example 5: geometric (p057-p059) - counts flips until first head

N = number of flips needed to get the first head

$$\begin{aligned} P(N=1) &= P(H) &&= p \\ P(N=2) &= P(TH) &&= (1-p)p \\ P(N=3) &= P(TTH) &&= (1-p)^2 p \\ \dots &&& \\ P(N=n) &= P(T\dots TH) &&= (1-p)^{n-1} p \end{aligned}$$



The sum verification (infinite geometric series, common ratio $(1-p)$):

$$\sum (1-p)^{n-1} p = p / (1 - (1-p)) = p/p = 1 \quad \checkmark$$

This is the GEOMETRIC distribution. Mean = $1/p$.
It has NO teaching slide for its mean, but the exam asks it (assignment 1 short Q4: $E = 2$).

7. Example 6: car lifetime (p060) - the bridge to continuous

X = lifetime of a car, assumed to take any value in an interval (a,b)
- not countable, cannot be listed as a sequence
-> NOT a discrete rv

8. Discrete vs continuous, the definitions (p061-p063)

DISCRETE	CONTINUOUS
finite OR countably infinite can be arranged as a sequence 1st, 2nd, 3rd, ...	all values in an interval CANNOT be listed as a sequence $P(X = \text{any single } c) = 0$

The formal continuous definition has TWO parts:

- possible values = all numbers in one interval (or a union of intervals)
e.g. $[0,10] \cup [20,30]$
- NO single value has positive probability: $P(X = c) = 0$

WHY $P(X=c)=0$ FOR CONTINUOUS
 $P(X=c) = \int \text{from } c \text{ to } c \text{ of } f(x) \, dx = 0$
 the width is zero, an area of zero width is zero
 -> endpoints never matter: $P(a < X < b) = P(a \leq X \leq b)$

9. The summary table to memorise

	DISCRETE	CONTINUOUS
support	list of values	interval(s)
tool	Σ (sum)	$\int$ (integral)
$P(X=x)$	$p(x) > 0$	0 always
$E(X)$	$\Sigma x \cdot p(x)$	$\int x \cdot f(x) \, dx$
$\text{Var}(X)$	$\Sigma (x-\mu)^2 p(x)$	$\int (x-\mu)^2 f(x) \, dx$

named examples	binomial, Poisson, geo	uniform, normal, exponential
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Next file builds the pmf and cdf (the two functions that describe any rv).